

Applied A.I. Solutions

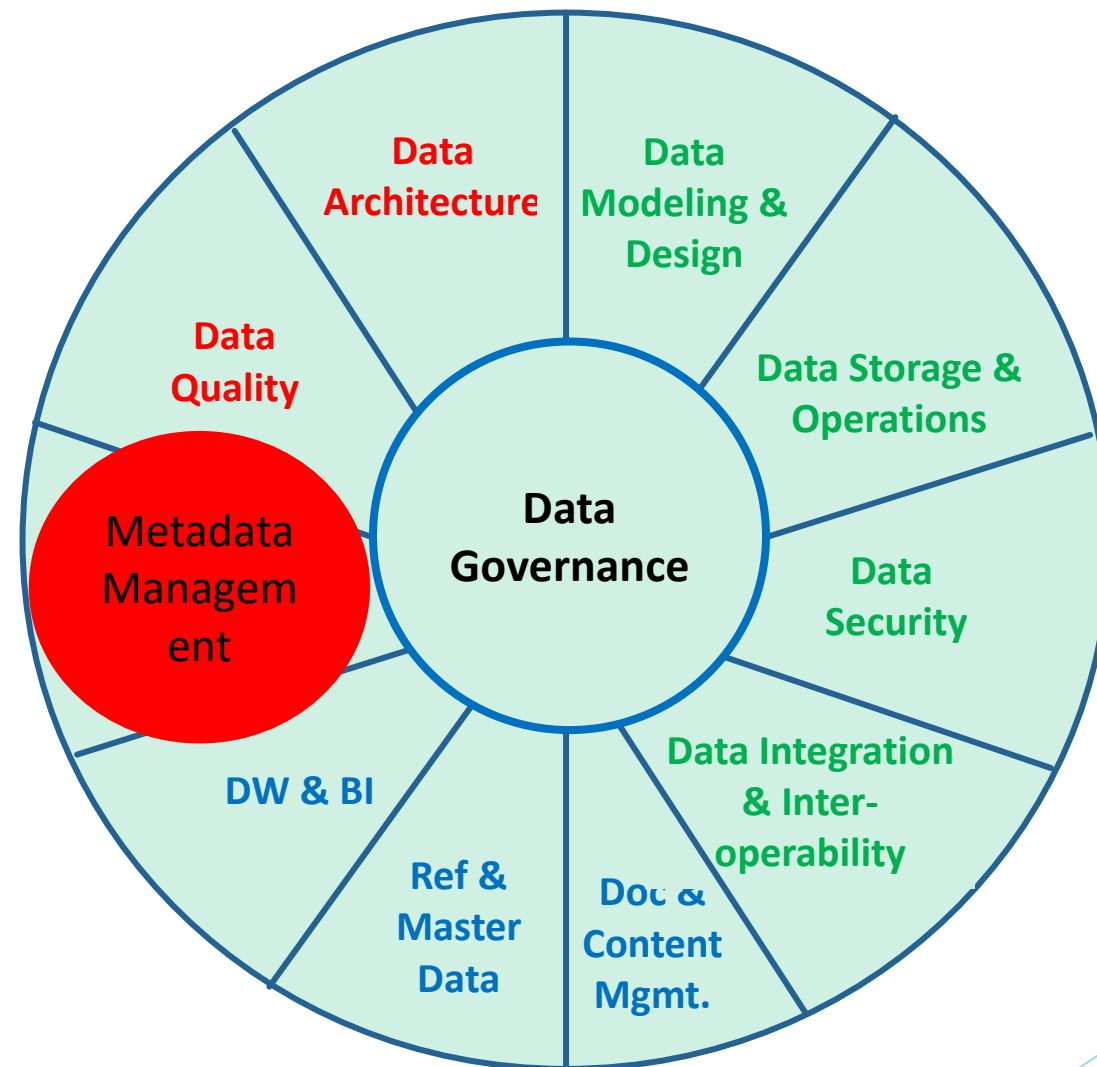
Foundations of Data Management

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METADATA MANAGEMENT

The DAMA Wheel



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INTRODUCTION

- Metadata is data about the asset (data)
- Metadata represents the context, information and knowledge about the data that we need to manage
- Metadata helps to interpret data in a meaningful way
- Metadata originates from a range of processes related to data creation, processing, and use, including architecture and governance

Metadata Management Framework

Definition

Planning, Implementation, and Control activities to enable access to high-quality, integrated metadata

Metadata Management programs are guided by the following principles:

- Organizational commitment
- Strategy
- Enterprise perspective
- Quality
- Improvement
- Audit
- Socialization
- Access

<https://www.gartner.com/doc/reprints?id=1-24L1M1LE&ct=201112&st=sb>

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Goals

1. Provide **organizational understanding of business terms and lineage**
2. Collect and integrate metadata from **diverse sources**
3. Provide a **standardized** way to access metadata
4. Ensure Metadata Quality and Security



Business Drivers

Others Business Drivers are ...

- Increase **confidence** in data
- Increase the **value** of strategic information
- Improve operational **efficiency**
- Improve **communication** between data consumers, IT professionals
- Prevent the use of out-to-date or **incorrect** data
- Create accurate impact analysis thus reducing **risks**
- Support regulatory **compliance**

Inputs

- Data architecture
- Business requirements
- Data governance metadata
- Metadata issues
- Business metadata
- Technical metadata
- Operational metadata



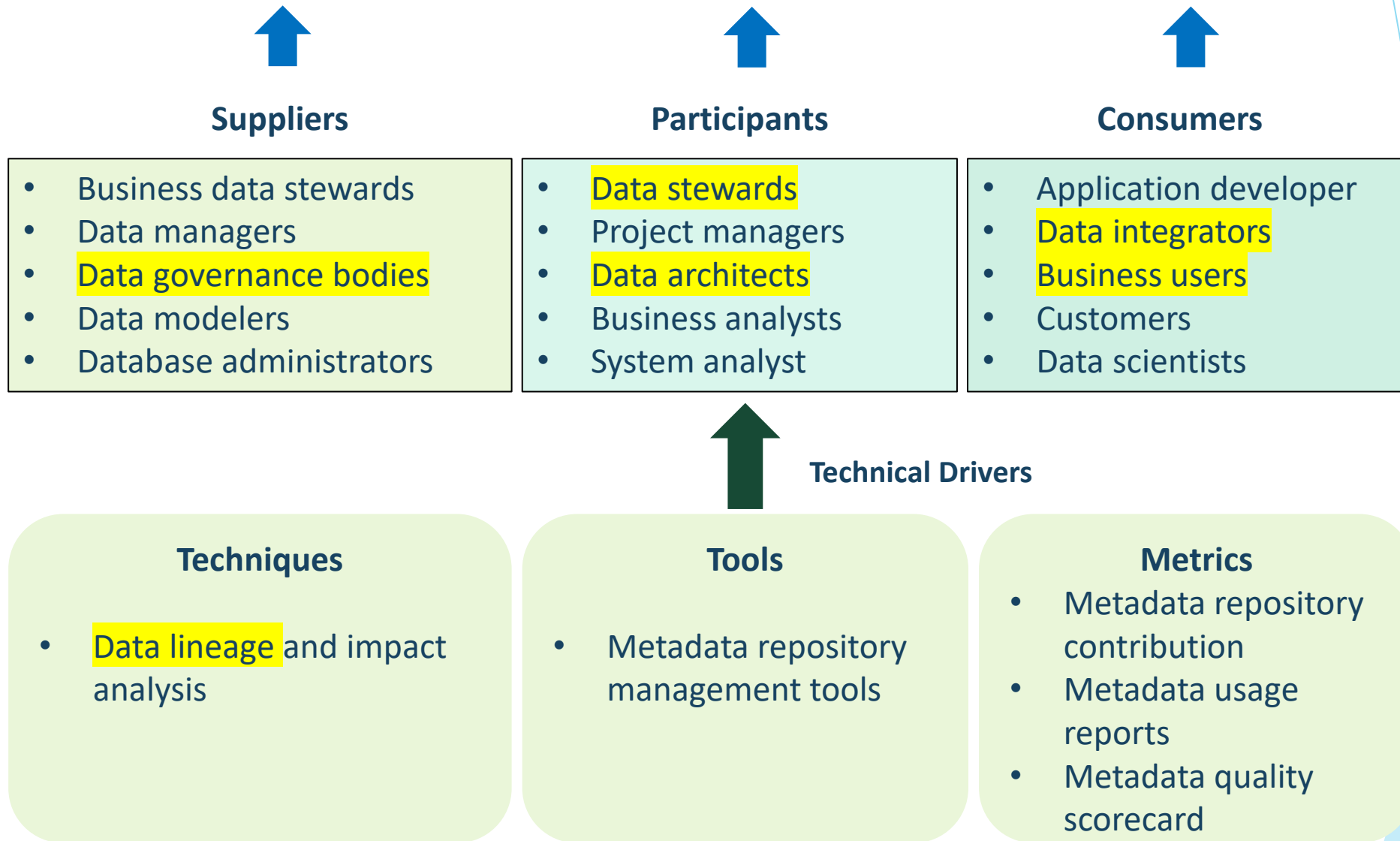
Activities

1. Define metadata strategy
2. Understand metadata requirements
3. Define Metadata architecture
 - Create metamodel
 - Apply metadata standards
 - Manage metadata stores
4. Create and maintain metadata
 - Integrate metadata
 - Distribute and deliver metadata
5. Query, report and analyze metadata



Deliverables

1. Metadata strategy
2. Metadata standards
3. Metadata architecture
 - a. Metamodel
 - b. Unified metadata
 - c. Data lineage
 - d. Impact analysis
 - e. Dependency analysis
 - f. Metadata control process



Essential Concepts

1. Metadata vs Data
2. Type of Metadata

Business Metadata

- Definitions, descriptions of data sets, tables and columns
- Business, transformation rules, calculations
- Data Models
- DQ rules, measurements
- Data update schedules

Technical Metadata

- Physical database table, column names, properties
- Access permissions
- Data CRUD
- Physical data models
- ETL job details
- File format schema definition

Operational Metadata

- Job execution logs (batch)
- History of extracts
- Schedule anomalies
- Audit results
- Error logs
- Reports, query access, frequency, execution time

Metadata Architecture

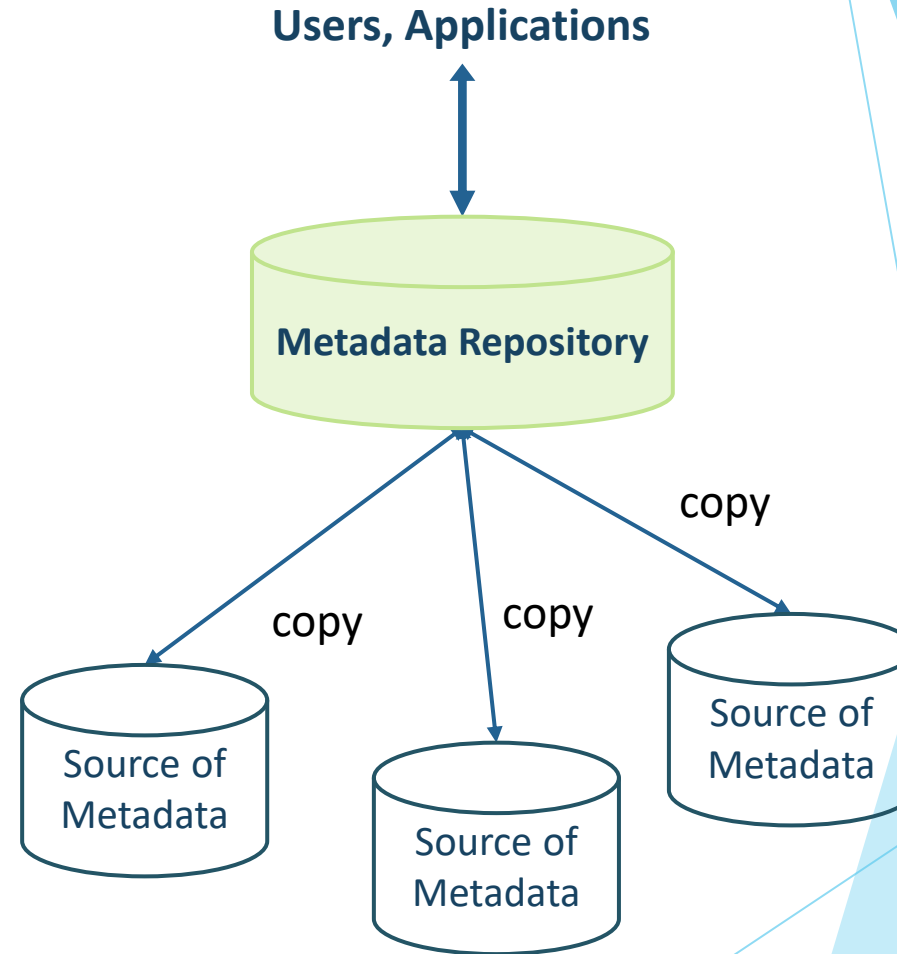
- **Centralized Metadata Architecture**
A single Metadata repository

Pros:

- High availability
- Quick Metadata retrieval
- Improved quality database structures

Cons:

- Complex processes
- Costly Maintenance



Metadata Architecture – cont'd

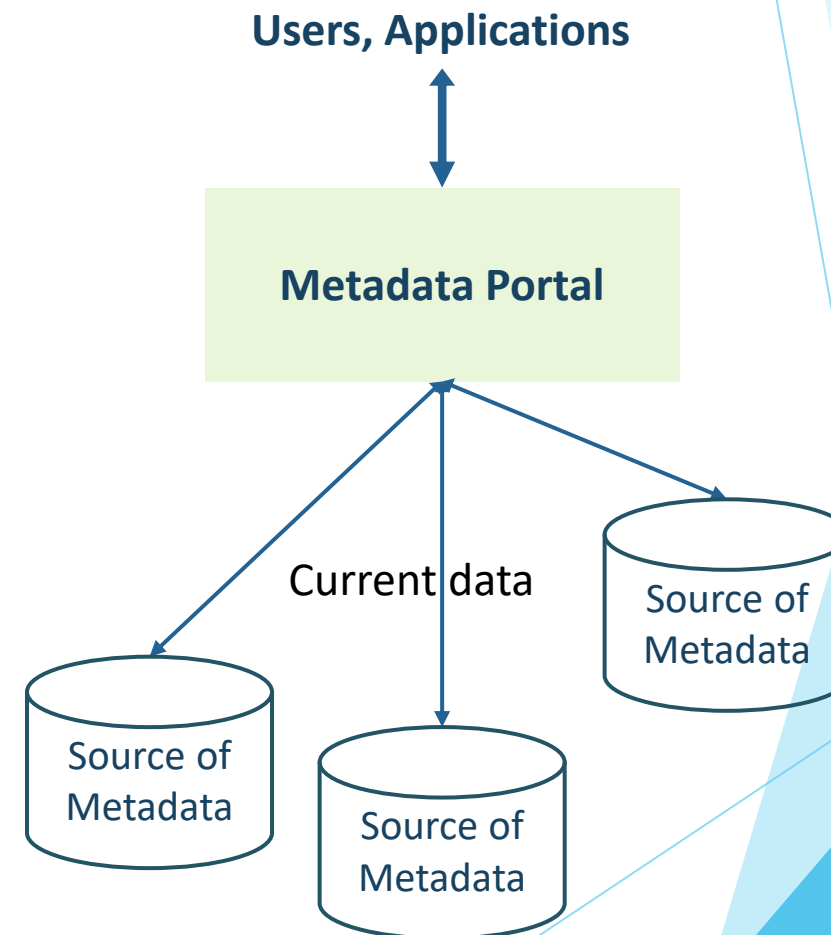
- **Distributed Metadata Architecture**
A single access point

Pros:

- Metadata always current, valid
- Queries are distributed, improving performance
- No metadata replication or synchronization processes

Cons:

- Query limitations by source systems availability
- Metadata quality depends on source systems



METADATA GOVERNANCE

- Process Controls
- Documentation of Metadata Solutions
- Metadata Standards and Guidelines
- Metrics
 - Metadata Management Maturity
 - Metadata repository availability and completeness
 - Metadata Usage and Business Glossary Activity
 - Metadata documentation quality

